TRUSTIA AI TECHNOLOGIES · HARDWARE & INTEGRATION PLAN

HYUNDAI IONIQ 5 LEVEL-4 ROBOTAXI PLAN

Date: October 31, 2026 | Version: v4.4 Enterprise (1,350 Tests Verified)

1. EXECUTIVE SUMMARY & OBJECTIVE

- Modular conversion of Hyundai Ioniq 5 E-GMP chassis into a Level-4 Robotaxi without drilling.
- Trustia AI C++20/ROS 2 autonomy core (1,350 automated tests, 100% green) connects directly to the vehicle CAN-FD bus, enabling sub-millisecond drive-by-wire control.

2. CORE ARCHITECTURE & SAFETY PROTOCOLS

- CAN-FD Direct Control: Native commanding of steering angle, acceleration, braking, and gears.
- 50 ms Watchdog Safety Timeout: Autonomous Minimal Risk Maneuver (MRM) fallback.
- 1.4 ms Bus Latency: 200x faster than human driver response times.
- 3.5 Nm Driver Takeover: Instant manual hand-off in under 5 ms upon steering wheel contact.

3. HARDWARE & SENSOR SUITE (BOM)

- 3D LiDAR: 128-Beam Ouster OS1-128 (2.6 million points/sec high-density spatial map).
- Deep Perception: CenterPoint BEV CNN Backbone (ONNX Runtime / TensorRT) · 25.4 ms latency.
- Localization: GNSS-Denied 400Hz Error-State Kalman Filter (ESKF) & 3D NDT Pose-Graph SLAM.
- Compute: Rugged, liquid-cooled, vibration-resistant industrial edge GPU/CPU unit.

4. PROVING GROUND COMMISSIONING ROADMAP (60 DAYS)

- Phase 1 (Days 1-15): Benchtop CAN-FD bus decoding, 60-ohm termination, and static actuation.
- Phase 2 (Days 16-35): Closed-track proving ground testing at 30 km/h with GPS jamming.
- Phase 3 (Days 36-60): Full 50 km/h urban speed dynamic obstacle avoidance and passenger pickup.